

22 Ans: C

Flux linkage in the current position = $NBA\sin\theta$
(i.e. horizontal component of field lines so they are perpendicular to the area in its current position)

Flux linkage when the coil is horizontal = $NBA\cos\theta$
(i.e. vertical component of field lines vertical so they are perpendicular to the area when horizontal)

Hence, the change in magnetic flux linkages would be $(NBA\cos\theta - NBA\sin\theta)$

21

22 **Ans: D**